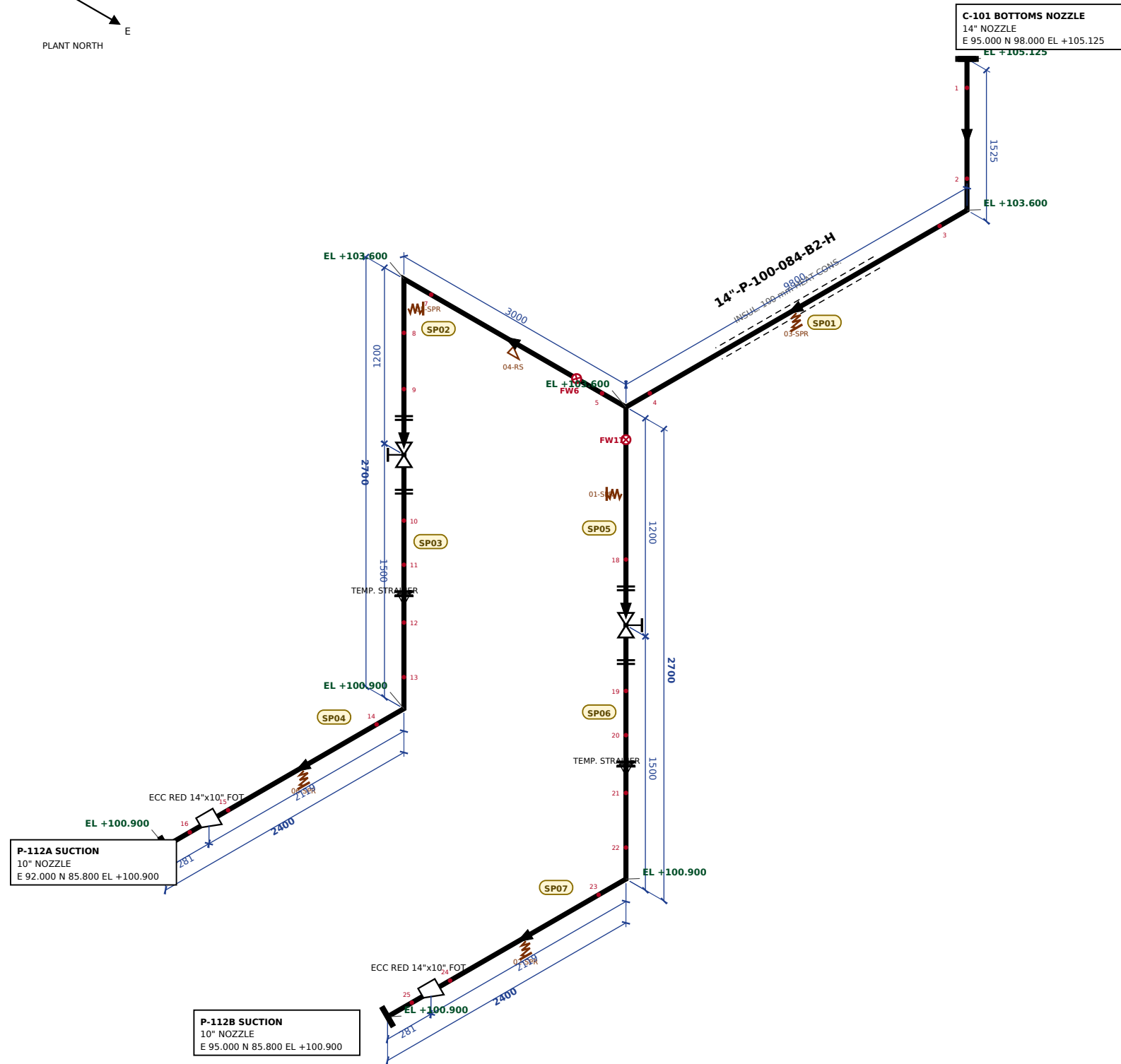
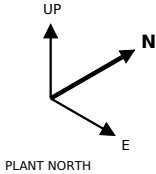


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore STD, ASTM A182 F5	14"	9
2	PIPE STD, BE, ASTM A335 P5 SMLS	14"	15.7 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WP5	14"	4
4	TEE EQUAL, BW, STD, ASTM A234 WP5	14"	1
5	REDUCER ECC, BW, STD, ASTM A234 WP5	14" x 10"	2
6	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	10"	2
7	ERECTION: GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	14"	2
8	TEMPORARY CONE STRAINER (commissioning)	14"	2
9	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	14"	9
10	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	14"	9
11	PIPE SUPPORTS (std. dwgs): rest x1, spring x5	-	6

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	14"	851	SP01-2	14"	8988	SP01-3	14"	350
SP02-1	14"	1838	SP02-2	14"	145	SP03-1	14"	37
SP04-1	14"	126	SP04-2	14"	1421	SP05-1	14"	399
SP06-1	14"	37	SP07-1	14"	126	SP07-2	14"	1421

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=12.0m SP02=3.2m SP03=0.3m SP04=3.2m SP05=0.5m SP06=0.3m SP07=3.2m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 6.6 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 14"x10" FLAT ON TOP; temporary strain
- Pump nozzle 10" assumed (API 610 vendor data pending).
- Line exits C-101 skirt through 1000 mm access sleeve; hot (390 C) - skirt opening insulated.
- NPSHa per mech. data sheet; suction line kept short (< 15 m developed per pump).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 14"-P-100-084-B2-H

C-101 BOTTOMS TO P-112A/B SUCTION

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			
CFU-100-PI-ISO-003			
REV			
A			